

# Detailed Connections Between Arduino and Seven Segment Display

## Aim

To interface a Seven Segment Display directly with Arduino Uno and display digits by driving the individual segments through Arduino digital output pins.

## 1. Components Used

| Component                            |
|--------------------------------------|
| Arduino Uno                          |
| Seven Segment Display (Common Anode) |
| Jumper Wires                         |
| Breadboard                           |

## 2. Purpose of Each Component

| Component             | Function                                       |
|-----------------------|------------------------------------------------|
| Arduino Uno           | Drives each segment HIGH/LOW to display digits |
| Seven Segment Display | Displays decimal digits 0-9                    |
| Jumper Wires          | Connects Arduino pins to display segment pins  |
| Breadboard            | Holds components and connections               |

## 3. Arduino to Seven Segment Display Connections

| Arduino Pin | Segment Pin | Segment          |
|-------------|-------------|------------------|
| D2          | Pin 7       | a (Top)          |
| D3          | Pin 6       | b (Top Right)    |
| D4          | Pin 4       | c (Bottom Right) |
| D5          | Pin 2       | d (Bottom)       |
| D6          | Pin 1       | e (Bottom Left)  |
| D7          | Pin 9       | f (Top Left)     |
| D8          | Pin 10      | g (Middle)       |

## 4. Power Connections

| Pin                          | Connection |
|------------------------------|------------|
| Pin 3 & Pin 8 (Common Anode) | +5V        |
| D5                           | GND        |

## 5. Seven Segment Display Pin Description

| Pin No. | Segment      | Position on Display   |
|---------|--------------|-----------------------|
| 1       | e            | Bottom Left vertical  |
| 2       | d            | Bottom horizontal     |
| 3 & 8   | Common Anode | +5V                   |
| 4       | c            | Bottom Right vertical |
| 5       | dp           | Decimal Point         |
| 6       | b            | Top Right vertical    |
| 7       | a            | Top horizontal        |
| 9       | f            | Top Left vertical     |
| 10      | g            | Middle horizontal     |

## 6. Overall Working

1. Arduino sets each segment pin HIGH or LOW.
2. Since Common Anode display: segment ON when pin is LOW, OFF when HIGH.
3. Arduino drives D2-D8 with correct pattern to display digits 0-9.
4. D5 connected to GND completes the circuit.

## 7. Data Flow of the System

Arduino Output Pins (D2-D8)

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Seven Segment Display Segments (a-g)

↓

Displayed Decimal Digit

## 8. Important Notes

- Use a Common Anode display only.
- Common Anode pins (3 & 8) must be connected to +5V.

- Ensure common GND between Arduino and display.
- For Common Anode: write LOW to turn a segment ON, HIGH to turn OFF.